SET-D

SOMAIYA VIDYAVIHAR UNIVERSITY

15.05.2025 (E)

Semester: January 2025– April 2025		
Maximum Marks: 50	Examination: End-Semester Examination	Duration: 2 Hrs. 30 min
Programme code: 05		
Programme: Honors Software and Mobile Application Development	Class: S.Y.	Semester: IV (SVU 2023)
Institute/School/Department: K. J. Somaiya School of Engineering	Name of the department: COMPUTER	
Course Code: 216H05C401	Name of the Course: Fundamental of Swift	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question Statement	Max. Marks
Q.1	Attempt any **four**	20
a	Define a closure? Write a closure that calculates the factorial of a number.	05
b	Explain an enum in Swift and how is it declared? Give one example.	05
c	Difference between var and let in Swift with examples. Why would you use one over the other?	05
d	Difference between an Array and a Set in Swift? Explain with example.	05
e	Explain the purpose of the guard statement? How is it different from if?	05
f	Explain the concept of Optionals in Swift. How is nil handled?	05
Q.2	Attempt any **One**	10
a	Implement a task manager considering following conditions: • Create a struct Task with a title (String) and isCompleted (Bool, default false). • Write a function that takes an array of Task and a task title. • The function should iterate through the array and set the isCompleted property to true for the task with the matching title. • Demonstrate creating an array of Task and marking one as completed. Print the isCompleted status of that task before and after.	
b	Develop a student management system that stores roll numbers as keys and names as values in a dictionary. • Write functions to add a new student, update a name, delete a student, and search by roll number. • Show how to handle cases where a key might not exist.	
Q.3	Attempt the following	20
	Create a Profile Settings screen consisting of the following elements: • A UIImageView to display profile picture.(Use System Icons) • UILabels and UITextFields for name, email, and phone number. • A UISwitch to indicate dark mode preference. • A UIButton to "Save Settings".	

	Implement: • Vertical StackViews for arrangement. • Constraints for adaptive UI. • Navigation segue to a confirmation screen with updated details.	